

Set

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Mathematical Form (1–5)

1. $\{1, 2, 3, 4, 5, 6\}$

$$X = \{x : x \in \mathbb{N}, 1 \leq x \leq 6\}$$

2. $\{2, 4, 6, 8, 10\}$

$$X = \{x : x \in \mathbb{N}, x \equiv 0 \pmod{2}, 2 \leq x \leq 10\}$$

3. $\{1, 3, 5, 7, 9\}$

$$X = \{x : x \in \mathbb{N}, x \equiv 1 \pmod{2}, 1 \leq x \leq 9\}$$

4. $\{5, 10, 15, 20, 25\}$

$$X = \{x : x = 5n, 1 \leq n \leq 5, n \in \mathbb{N}\}$$

5. $\{1, 4, 9, 16, 25\}$

$$X = \{x : x = n^2, 1 \leq n \leq 5, n \in \mathbb{N}\}$$

Mathematical Form (6–10)

6. $\{1, 8, 27, 64, 125\}$

$$X = \{x : x = n^3, 1 \leq n \leq 5, n \in \mathbb{N}\}$$

7. $\{2, 3, 5, 7, 11\}$

$$X = \{x : x \text{ is prime}, 2 \leq x \leq 11\}$$

8. $\{10, 20, 30, 40, 50\}$

$$X = \{x : x = 10n, 1 \leq n \leq 5, n \in \mathbb{N}\}$$

9. $\{3, 6, 9, 12, 15\}$

$$X = \{x : x = 3n, 1 \leq n \leq 5, n \in \mathbb{N}\}$$

10. $\{0, 1, 2, 3, 4, 5\}$

$$X = \{x : x \in \mathbb{Z}, 0 \leq x \leq 5\}$$

General Inclusion–Exclusion Principle

For finite sets $A_1, A_2, \dots, A_n$,

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| &= \sum_{i=1}^n |A_i| \\ &\quad - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \\ &\quad + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| \\ &\quad - \dots \\ &\quad + (-1)^{n-1} |A_1 \cap A_2 \cap \dots \cap A_n|. \end{aligned}$$

Cardinality of a Set & Finite Set

Cardinality of a Set

The cardinality of a set is the number of distinct elements in the set. It is denoted by $|A|$.

Example

$$A = \{\text{Python}, \text{R}, \text{SQL}, \text{TensorFlow}\}$$

$$|A| = 4$$

Finite Set

A finite set contains a limited number of elements.

$$B = \{\text{CNN}, \text{RNN}, \text{Transformer}, \text{GAN}\}$$

Since

$$|B| = 4,$$

B is a finite set.

Infinite Set & Null Set

Infinite Set

A set containing infinitely many elements.

Example

All possible values of neural network weights

$$W = \{w : w \in \mathbb{R}\}$$

Hence,

$$|W| = \infty$$

Null (Empty) Set

A set containing no elements.

Notation:

$$\emptyset$$

Example:

Suppose no student completed an AI Ethics certification.

Equal Set

Definition

Two sets are equal if they contain exactly the same elements.

Example

$$A = \{\text{Python}, \text{SQL}, \text{Java}\}$$

$$B = \{\text{Java}, \text{Python}, \text{SQL}\}$$

$$A = B$$

Relations

Definition

A relation from a set A to a set B is a subset of the Cartesian product $A \times B$.

$$R \subseteq A \times B$$

If $(a, b) \in R$, then a is related to b , written as aRb .

Example:

Let

$$A = \{\text{CNN, LSTM, Transformer}\}$$

$$B = \{\text{Image, Sequence, Text}\}$$

Define

$$R = \{(\text{CNN, Image}), \\ (\text{LSTM, Sequence}), \\ (\text{Transformer, Text})\}.$$

Since $R \subseteq A \times B$, R is a relation from A to B .

Equivalence Relation

Definition

A relation R on a set S is an equivalence relation if it is:

Reflexive, Symmetric, Transitive

Example

$$S = \{\text{CNN}, \text{ResNet}, \text{LSTM}, \text{GRU}\}$$

Define

$$xRy \iff x \text{ and } y \text{ belong to the same model family.}$$

Examples:

$$\text{CNN } R \text{ ResNet}, \quad \text{LSTM } R \text{ GRU}$$

Hence, R is an equivalence relation.

Subset

Definition

A set A is called a subset of B if every element of A is also an element of B .

Notation:

$$A \subseteq B$$

Example

$$A = \{\text{Python}, \text{SQL}\}$$

$$B = \{\text{Python}, \text{SQL}, \text{Java}, \text{R}\}$$

$$A \subseteq B$$

Universal Set

Definition

The universal set contains all objects under consideration.

Notation:

$$U$$

Example

$$U = \{\text{Python, Java, C++}, \text{SQL, R}\}$$

Superset

Definition

A set B is called a superset of A if every element of A belongs to B .

Notation:

$$B \supseteq A$$

Example

$$A = \{\text{Python}, \text{SQL}\}$$

$$B = \{\text{Python}, \text{SQL}, \text{Java}\}$$

$$B \supseteq A$$

Power Set

Definition

The power set is the set of all subsets of a set.

Notation:

$$P(A)$$

If

$$|A| = n,$$

then

$$|P(A)| = 2^n$$

Example

$$A = \{\text{Python}, \text{SQL}\}$$

$$P(A) = \{\emptyset, \{\text{Python}\}, \{\text{SQL}\}, \{\text{Python}, \text{SQL}\}\}$$

Singleton Set

Definition

A singleton set contains exactly one element.

Example

$$A = \{\text{TensorFlow}\}$$

Since

$$|A| = 1,$$

A is a singleton set.

Union

Definition

The union of two sets contains all elements that belong to either set.

Notation:

$$A \cup B$$

Example

$$A = \{\text{Python}, \text{R}\}$$

$$B = \{\text{SQL}, \text{Python}\}$$

$$A \cup B = \{\text{Python}, \text{R}, \text{SQL}\}$$

Intersection

Definition

The intersection of two sets contains the common elements of both sets.

Notation:

$$A \cap B$$

Example

$$A = \{\text{Python, SQL, Java}\}$$

$$B = \{\text{Python, R, SQL}\}$$

$$A \cap B = \{\text{Python, SQL}\}$$

Set Difference

Definition

The set difference contains elements that are in A but not in B .

Notation:

$$A - B$$

Example

$$A = \{\text{Python, SQL, Java}\}$$

$$B = \{\text{Python, R}\}$$

$$A - B = \{\text{SQL, Java}\}$$

Symmetric Difference

Definition

The symmetric difference contains elements that belong to exactly one of the two sets.

Notation:

$$A \triangle B$$

Formula:

$$A \triangle B = (A - B) \cup (B - A)$$

Example

$$A = \{\text{Python}, \text{SQL}\}$$

$$B = \{\text{Python}, \text{R}\}$$

$$A \triangle B = \{\text{SQL}, \text{R}\}$$

Complement

Definition

The complement of a set contains all elements in the universal set that are not in the set.

Notation:

$$A^c$$

Example

$$U = \{\text{Python, Java, SQL, R}\}$$

$$A = \{\text{Python, SQL}\}$$

$$A^c = \{\text{Java, R}\}$$

Cartesian Product

Definition

The Cartesian product is the set of all ordered pairs formed by taking one element from each set.

Notation:

$$A \times B$$

Example

$$A = \{\text{CNN}, \text{RNN}\}$$

$$B = \{\text{TensorFlow}, \text{PyTorch}\}$$

$$A \times B = \left\{ \begin{array}{l} (\text{CNN}, \text{TensorFlow}), \\ (\text{CNN}, \text{PyTorch}), \\ (\text{RNN}, \text{TensorFlow}), \\ (\text{RNN}, \text{PyTorch}) \end{array} \right\}$$

Partially Ordered Set (Poset)

Definition

A relation R on a set A is called a **partially ordered relation** if it satisfies:

- ▶ Reflexive: $aRa, \forall a \in A$
- ▶ Antisymmetric: If aRb and bRa , then $a = b$
- ▶ Transitive: If aRb and bRc , then aRc

The pair (A, R) is called a **Partially Ordered Set (Poset)**.

Example:

The relation "divides" ($|$) on the set

$$A = \{1, 2, 3, 6\}$$

is a partially ordered relation.

Example 1: Divisibility Relation

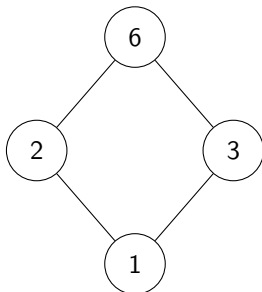
Let

$$A = \{1, 2, 3, 6\}$$

Define

$$aRb \iff a \mid b$$

where " $\mid$ " means "divides".



Hasse Diagram: The diagram shows the divisibility order after removing reflexive and transitive edges.

Example 2: Subset Relation

Let

$$A = \{a, b\}$$

The power set of A is

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

Define the relation

$$XRY \iff X \subseteq Y$$

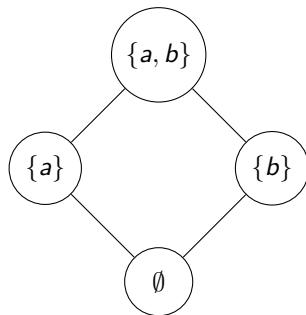
where " $\subseteq$ " means "is a subset of".

Since the subset relation is reflexive, antisymmetric and transitive,

$$(P(A), \subseteq)$$

is a partially ordered set.

Hasse Diagram for $(P(A), \subseteq)$



Observation

The Hasse diagram represents the subset ordering of the power set. Only the covering relations are shown.

Example 3: AI Model Hierarchy

Let

$$A = \{\text{AI, Machine Learning, Deep Learning, CNN, RNN}\}$$

Define the relation

$$xRy \iff x \text{ is a subtype of } y.$$

Examples:

CNN R Deep Learning

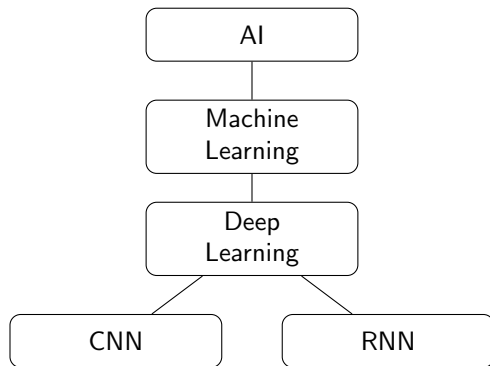
RNN R Deep Learning

Deep Learning R Machine Learning

Machine Learning R AI

Hence, (A, R) is a partially ordered set.

Hasse Diagram for AI Model Hierarchy



Observation

The diagram shows the hierarchy of AI models. CNN and RNN are incomparable, but both are covered by Deep Learning.

Chains and Antichains

Chain

A chain is a subset of a poset in which every pair of elements is comparable.

Example:

$$\text{CNN} < \text{Deep Learning} < \text{Machine Learning} < \text{AI}$$
$$\{\text{CNN}, \text{Deep Learning}, \text{Machine Learning}, \text{AI}\}$$

is a chain.

Antichain

An antichain is a subset of a poset in which no two distinct elements are comparable.

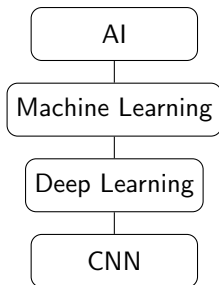
Example:

CNN and RNN are both deep learning models, but neither is a subtype of the other.

$$\{\text{CNN}, \text{RNN}\}$$

is an antichain.

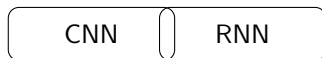
Chain and Antichain – Hasse Diagram



Chain

Every pair is comparable.

$$CNN < DL < ML < AI$$



Antichain

No two distinct elements are comparable.

$$\{CNN, RNN\}$$